

Computer Science & IT

Theory of Computation



Context Free Languages

Lecture No. 6



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Topics to be Covered



Topic

Last lecture of CFG

Topic

From Tomorrow Turning m/c

Topic

Ambiguity of grammarst

w $\in L$

w $\rightarrow$ more than one parse tree

$\Leftrightarrow$ more than one LMD

$\Leftrightarrow$ more than one RMD

Rule ①

Rule ②

Rule ③

Ambiguity is undecidable

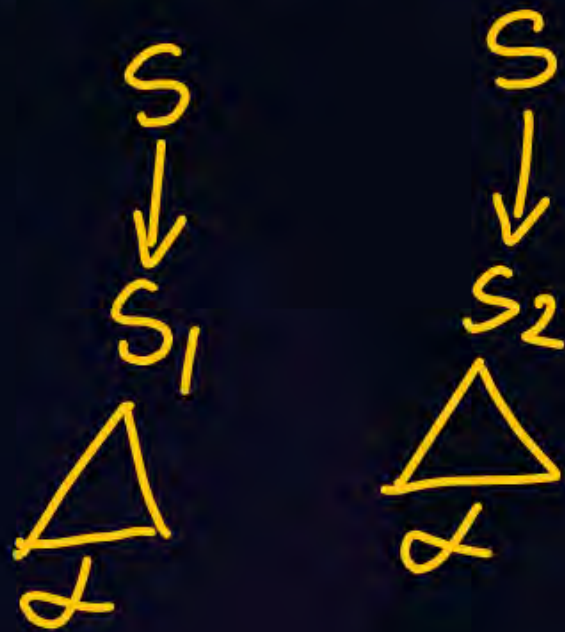
↓
no algorithm ✓



Ambiguity of grammars:-

Rule ①:- If there are productions like $S \rightarrow S_1 / S_2$ and if then there are two path $\begin{matrix} S \\ \downarrow \\ S_1 \end{matrix}$ $\begin{matrix} S \\ \downarrow \\ S_2 \end{matrix}$

$L(S_1) \cap L(S_2) \neq \phi$, then grammar is ambiguous



E/L: $S_1 \xRightarrow{*} \alpha$
 $S_2 \xRightarrow{*} \alpha$

$$\begin{array}{l}
 S \rightarrow \underline{S_1} / \underline{S_2} \\
 S_1 \rightarrow a \underline{S_1} / \underline{\epsilon} \\
 S_2 \rightarrow b \underline{S_2} / \underline{\epsilon}
 \end{array}
 \quad
 \begin{array}{l}
 S \rightarrow S_1 / S_2 / \epsilon \\
 S_1 \rightarrow a S_1 / a \\
 S_2 \rightarrow b S_2 / b
 \end{array}$$

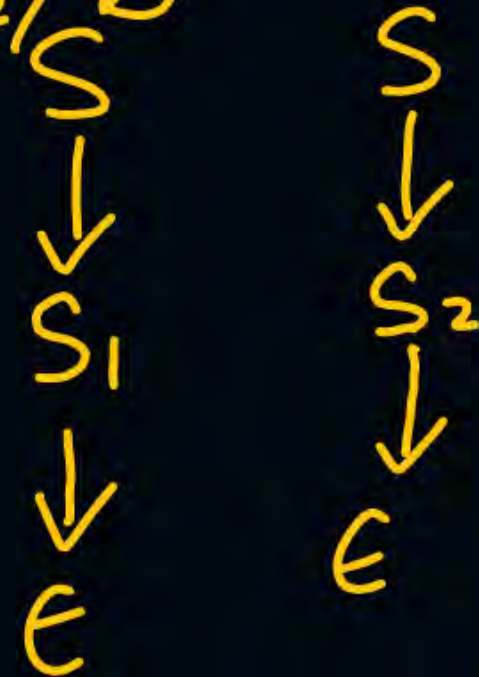
a) ambi° b) unambi°

$$S_1 \xRightarrow{*} a^*$$

$$S_2 \xRightarrow{*} b^*$$

$$a^* \cap b^* = \{\epsilon\}$$

For ϵ , we got two parse trees, so grammar is ambi°



unambiguous

$$S \rightarrow \underline{S_1} / \underline{S_2}$$

$$S_1 \rightarrow a \underline{S_1} / \underline{\epsilon}$$

$$S_2 \rightarrow b \underline{S_2} / \underline{\epsilon}$$

$$S_1 \stackrel{*}{\Rightarrow} a^*$$

$$S_2 \stackrel{*}{\Rightarrow} b^*$$

$$S = S_1 \cup S_2 \\ = a^* \cup b^* \checkmark$$

$$S \rightarrow S_1 / S_2 / \epsilon$$

$$S_1 \rightarrow a S_1 / a$$

$$S_2 \rightarrow b S_2 / b$$

$$S_1 \stackrel{*}{\Rightarrow} a^+$$

$$S_2 \stackrel{*}{\Rightarrow} b^+$$

$$S \rightarrow S_1 \cup S_2 \cup \{\epsilon\}$$

$$a^+ \cup b^+ \cup \{\epsilon\}$$

$$\underline{\underline{a^* \cup b^*}}$$

Sometimes we can convert ambiguous grammars to unambiguous grammars, but not always

$$S \rightarrow \underline{A}B/\underline{C}D$$

$$A \rightarrow aA/\epsilon$$

$$B \rightarrow bbB/\epsilon$$

$$C \rightarrow cC/\epsilon$$

$$D \rightarrow ddD/\epsilon$$

$$A^* \Rightarrow a^*$$

$$B^* \Rightarrow (bb)^*$$

$$AB^* \Rightarrow a^*(bb)^*$$

$$C^* \Rightarrow c^*$$

$$D^* \Rightarrow (dd)^*$$

$$CD \Rightarrow c^*(dd)^*$$



a) ambli

b) unambli

$$(AB) \cap (CD) = a^*(bb)^* \cap c^*(dd)^* = \{\epsilon\}$$

$$S \rightarrow AB/CD$$

$$A \rightarrow aA/\epsilon$$

$$B \rightarrow bbB/\epsilon$$

$$C \rightarrow cC/c$$

$$D \rightarrow ddD/\epsilon$$

$$A^* \Rightarrow a^*$$

$$B^* \Rightarrow (bb)^*$$

$$C^* \Rightarrow c^+$$

$$D \Rightarrow (dd)^*$$

$$AB \Rightarrow a^*(bb)^*$$

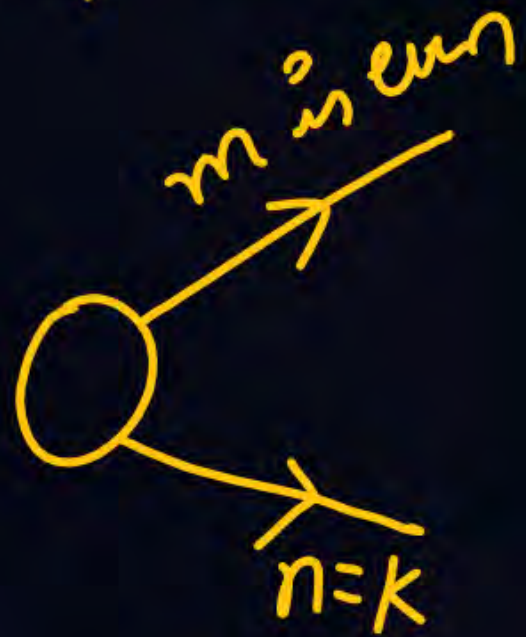
$$CD \Rightarrow c^+(dd)^*$$

$$AB \cap CD = a^*(bb)^* \cap c^+(dd)^*$$

$$= \phi$$

a) ambi b) unambi

$a^m b^n c^k / m = \text{even}, \Delta n = k$



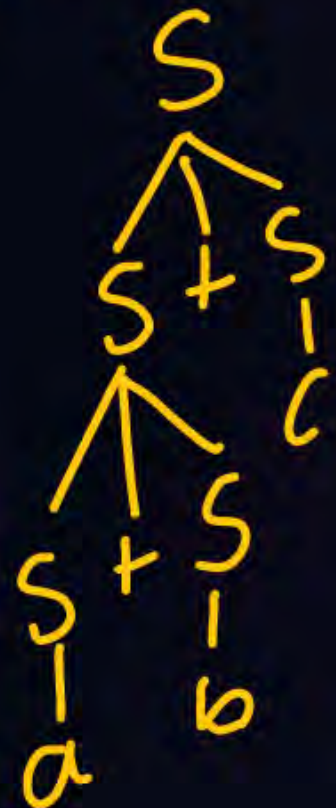
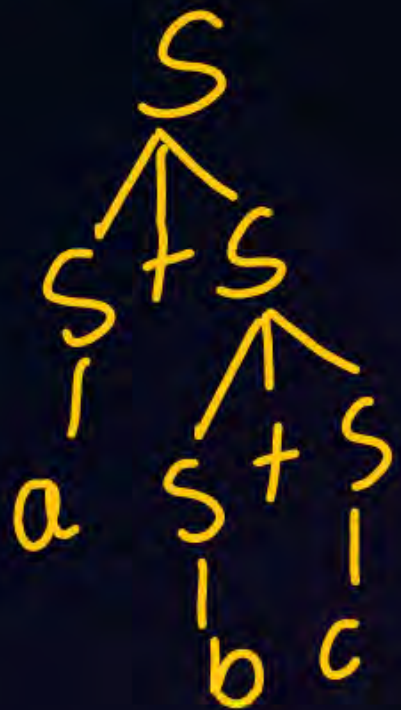
NPDA

Rule 2:-

Left recursion and non determinism may lead to ambiguity
 $\downarrow$
may Δ may not

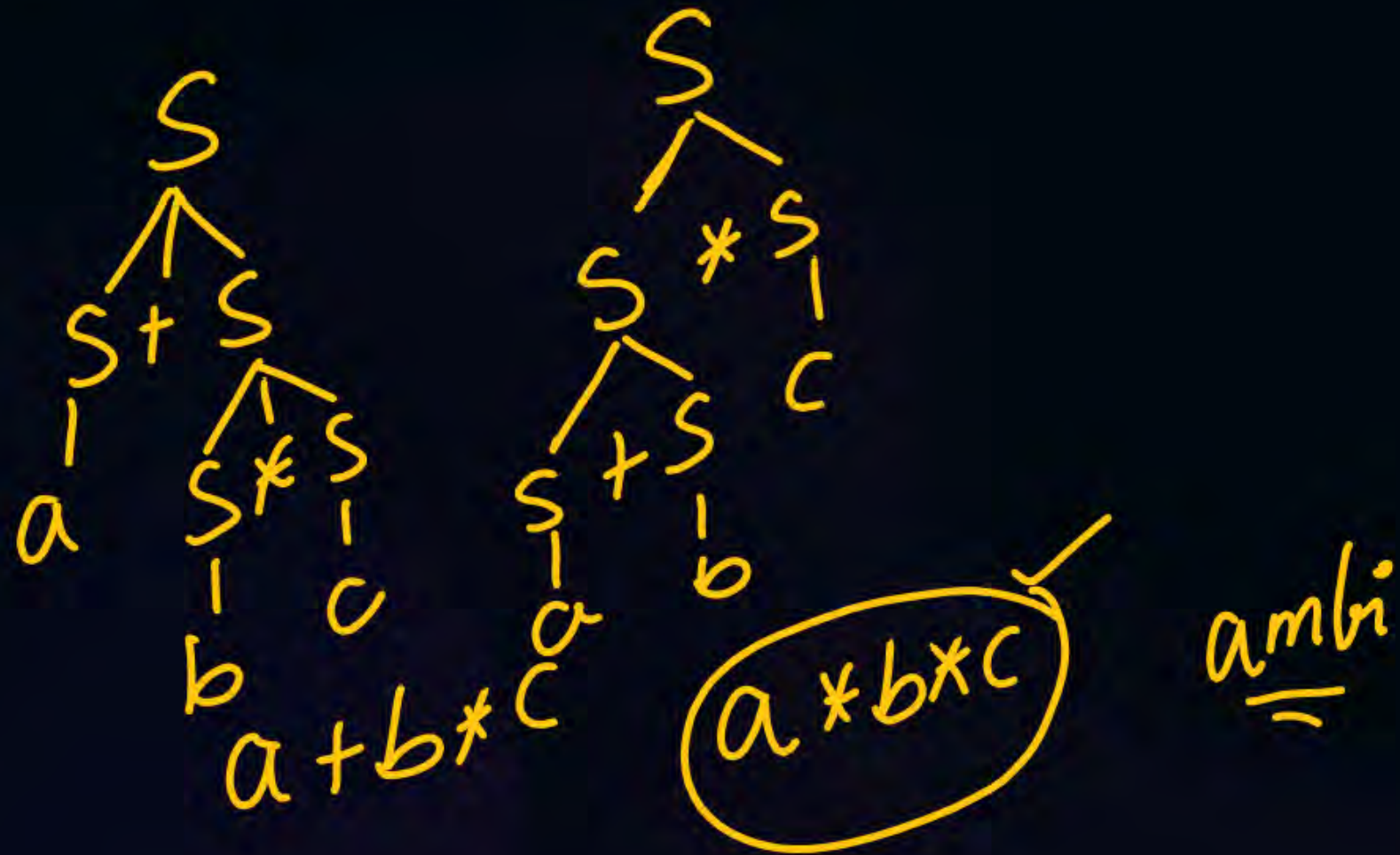
$S \rightarrow S + S / a / b / c \rightarrow (CD) \checkmark$

a) ambi b) unambi?



$a + b + c$
 $\Rightarrow$ ambiguous

$$S \rightarrow \underline{S+S} / \underline{S*S} / a / b / c$$



$$S \rightarrow S a / \epsilon$$



Left recursive

unambiguous

LR may not always lead to ambiguity

non deterministic grammars:-

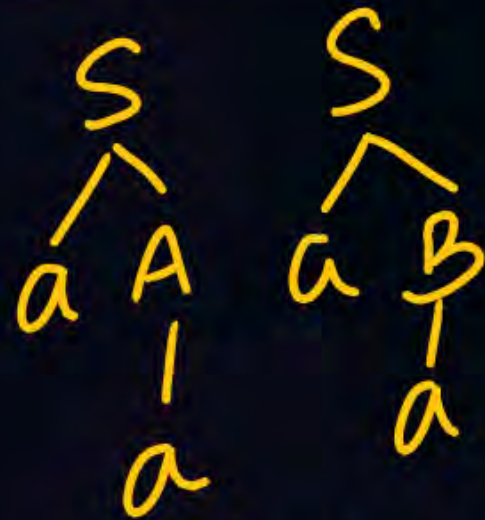
$$S \rightarrow \underline{\alpha} \underline{\gamma} / \underline{\alpha} \underline{\gamma} \checkmark$$

$$\underline{L(x) \cap L(y) \neq \phi}$$

$$S \rightarrow \underline{a} \underline{A} / \underline{a} \underline{B}$$

$$A \rightarrow a$$

$$B \rightarrow a$$



aa $\Rightarrow$ we got 2 parse trees.

$$S \rightarrow aA / aB$$

$$A \rightarrow aA / \epsilon$$

$$B \rightarrow bB / \epsilon$$

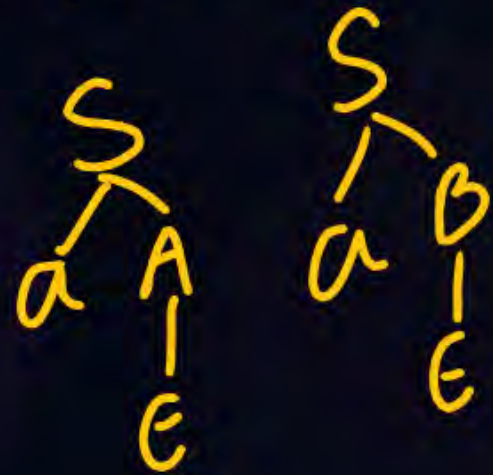
$$A \xRightarrow{*} a^*$$

$$B \xRightarrow{*} b^*$$

$$L(A) \cap L(B) \neq \emptyset$$

$$L(A) \cap L(B) = \epsilon$$

a) ambi b) unambi



a → Two parse tree

Rule:- Presense of SS

$$S \rightarrow a_b / _b a / SS / \epsilon$$

a)ambi b)unambi

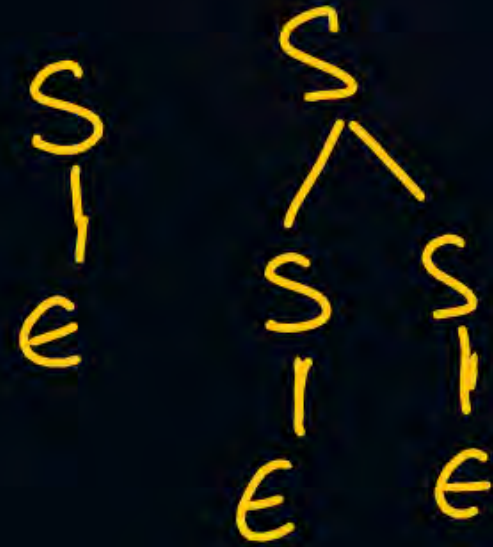
$$(ab+ba)^*$$

$$S \rightarrow \underline{a} / \underline{c} / \underline{SS} / \underline{\epsilon}$$

$$\& (a+c)^*$$

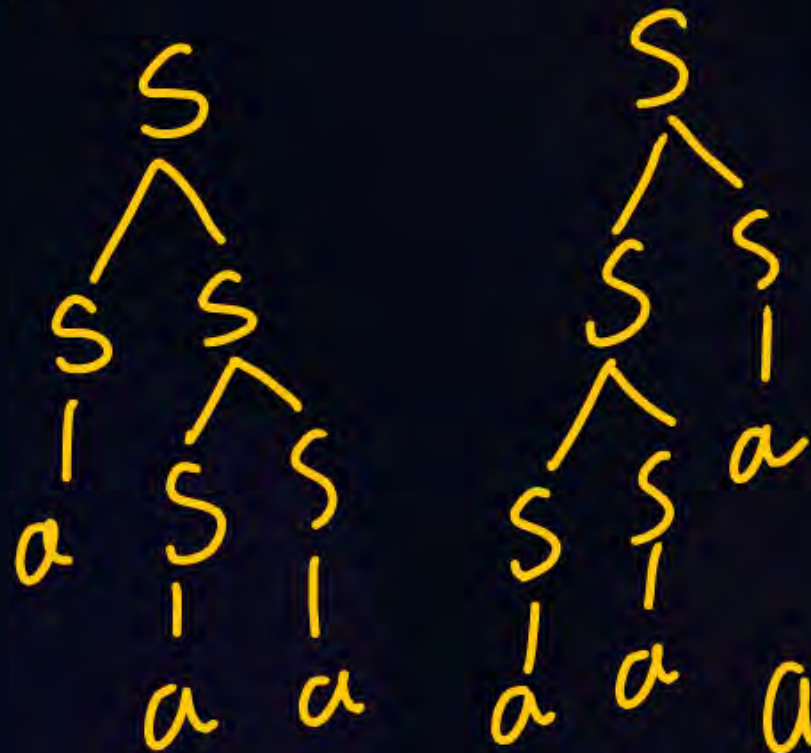
$$S \rightarrow 01 / 10 / SS / \epsilon$$

$$(01+10)^*$$



$S \rightarrow ab/ba/ss/a$

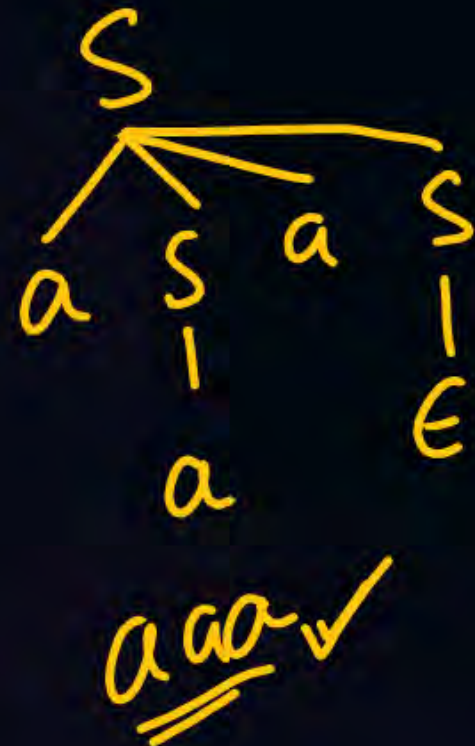
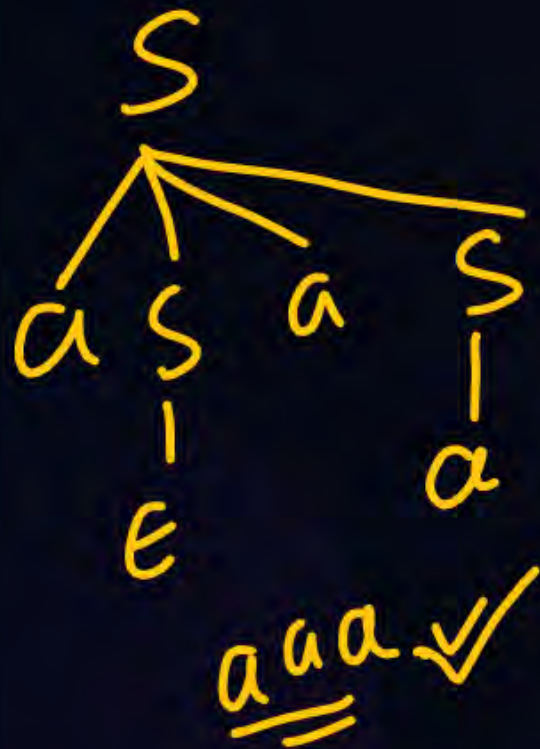
a) amb b) unamb



$aaa \Rightarrow 2$ parse trees

$S \rightarrow aSaS/a/\epsilon$

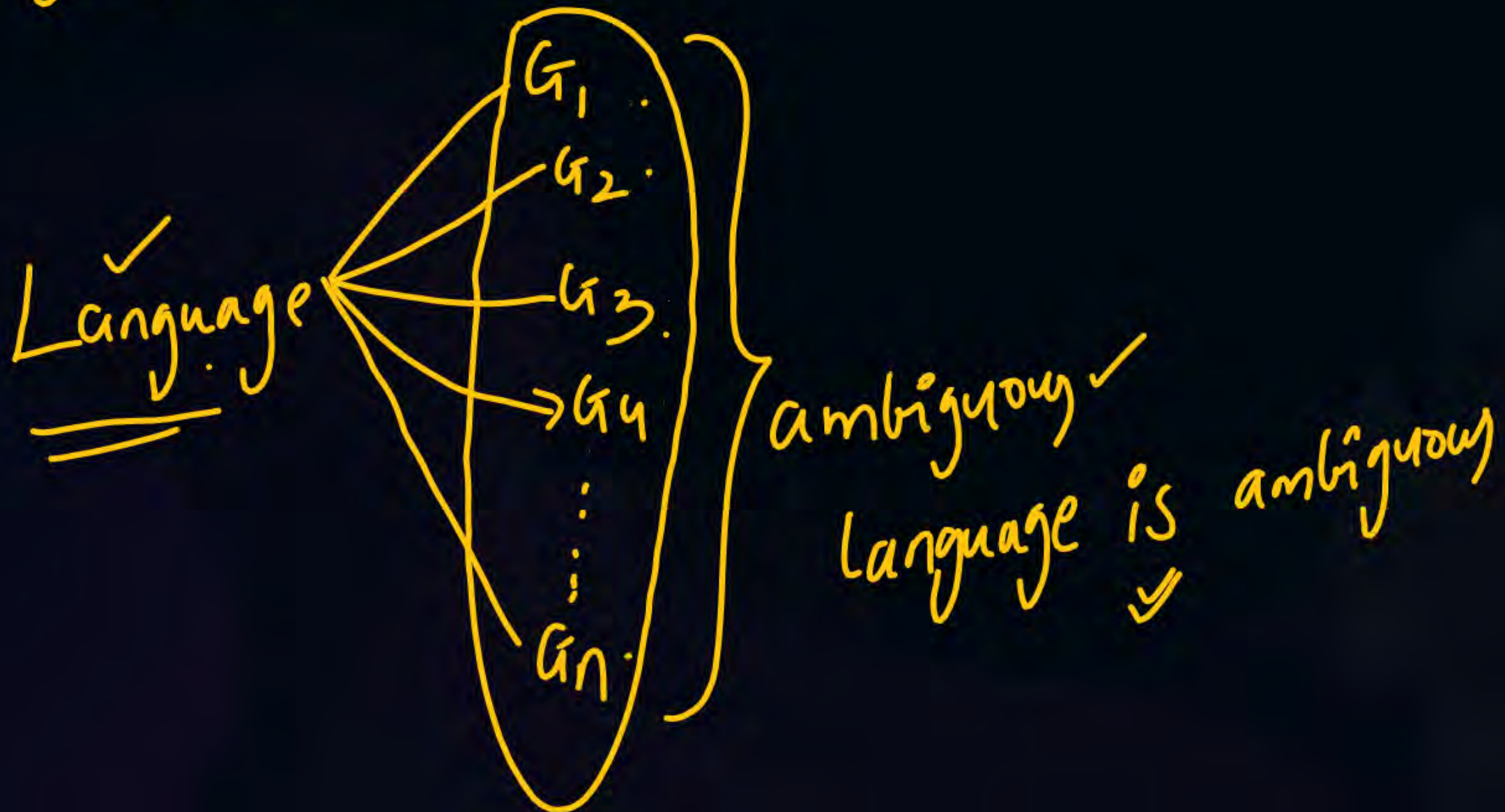
a) ambi b) unambi



ambi ✓

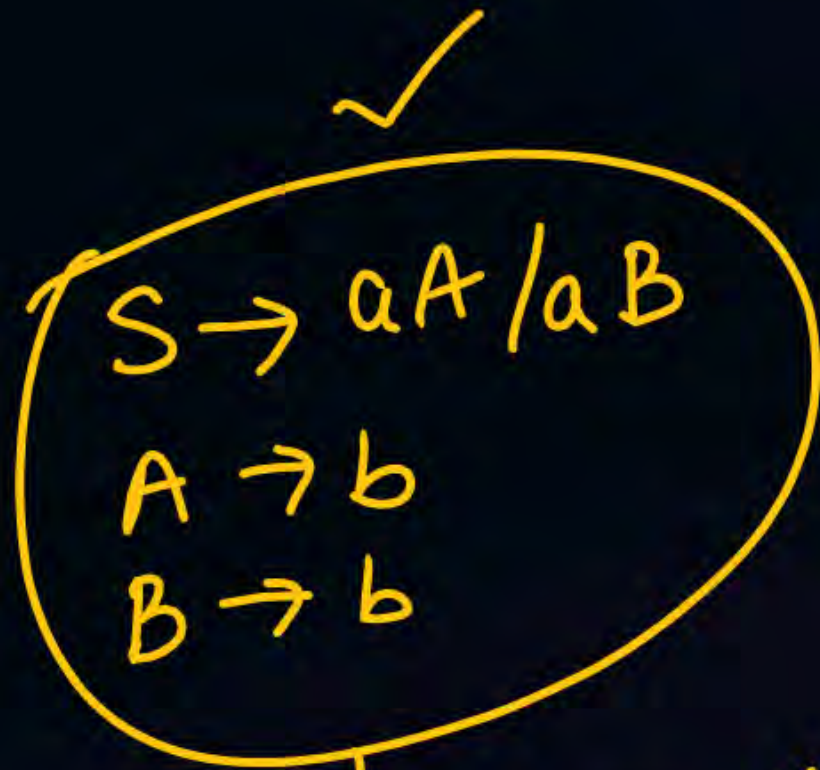
ambiguity of grammars ✓✓

ambiguity of languages ✓✓



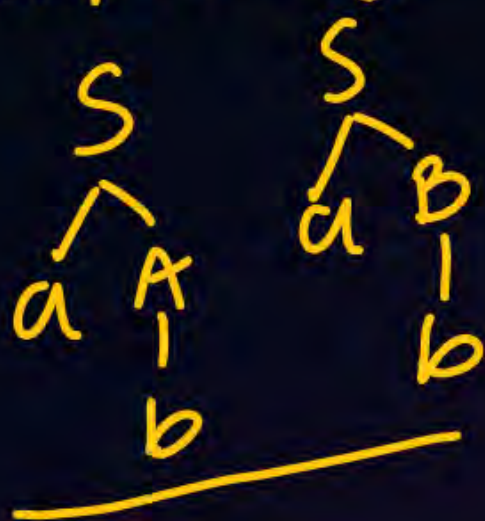
If all grammars of a language are ambiguous
then the language is said to be ambiguous.

Ambiguous languages are also called inherently ambiguous
languages



↓
unambiguous

grammar ambiguous ✓
 $L = \{ab\}$



If for a language, there is unambiguous grammar, then language is unambiguous

To check if a language is ambiguous

Rule 1:- L should be union of 2 or more languages ✓

$$L = \underline{L_1 \cup L_2} \quad \Delta \quad L = L_1 \cup L_2 \cup L_3 \quad \Delta \quad L = L_1 \cup L_2 \cup L_3 \cup L_4 \dots$$

Rule 2:- $L_1 \cup L_2$ should not collapse into a single language ✓

$$\underline{L = a^n b^n \cup a^* b^*}$$

collapse
 $a^* b^*$ ✓

Rule 3:- $L_1 \cap L_2 \neq \phi$
 $L_1 \cap L_2$ is infinite set



$$L = L_1 \cup L_2$$

$$S \rightarrow S_1 / S_2 / \epsilon, \quad S_1 \xRightarrow{*} L_1$$

$$S_2 \xRightarrow{*} L_2$$

$$\underline{\underline{L_1 \cap L_2}} = \underline{\underline{\emptyset}} = \checkmark \checkmark$$

$$\begin{array}{c} S \\ | \\ S_1 \\ \wedge \end{array} \quad \begin{array}{c} S \\ | \\ S_2 \\ \wedge \end{array}$$

$$L = \underline{a^n b^n} \cup \underline{a^n b^{2n}} / (n \geq 0) \quad n \geq 1$$

$$1) L = \underline{L_1 \cup L_2}$$

$$2) \underline{L_1 \cup L_2} \quad \times \text{collapse}$$

$$\times 3) \underline{L_1 \cap L_2} = \{\epsilon\} \rightarrow \text{finite} \checkmark \Rightarrow L \text{ is } \underline{\text{unambiguous}}$$

$$\begin{aligned} S &\rightarrow A/B \\ A &\rightarrow aAb/\epsilon \\ B &\rightarrow aBbb/\epsilon \end{aligned}$$

$$\begin{array}{cc} S & S \\ | & | \\ A & B \\ | & | \\ \epsilon & \epsilon \end{array}$$

$$\begin{aligned} S &\rightarrow A/B/\epsilon \\ A &\rightarrow aAb/ab \\ B &\rightarrow aBbb/\underline{abb} \end{aligned}$$

→ unambig grammar

$$L = a^n b^n \cup a^{2n} b^{2n} / n \geq 0$$

a) ambli b) unambli

1) $L = L_1 \cup L_2$

2) should not collapse

$$a^n b^n \cup \underline{a^{2n} b^{2n}} = \underline{a^n b^n}^{\checkmark}$$

unambiguous lang

$$L = \underline{a^n b^n} c^m \cup a^n \underline{b^m c^m} / n \geq 0$$

a) ambi b) unambi

Rule 1:- $L = L_1 \cup L_2$

Rule 2:- collapse x

Rule 3:- $\underline{a^n b^n} c^m \cap a^n \underline{b^m c^m}$
 $\underline{a^n b^n} c^n \rightarrow \infty \text{ set}$

ambiguous

$$S \rightarrow S_1 / S_2$$

$$\checkmark S_1 \rightarrow AC$$

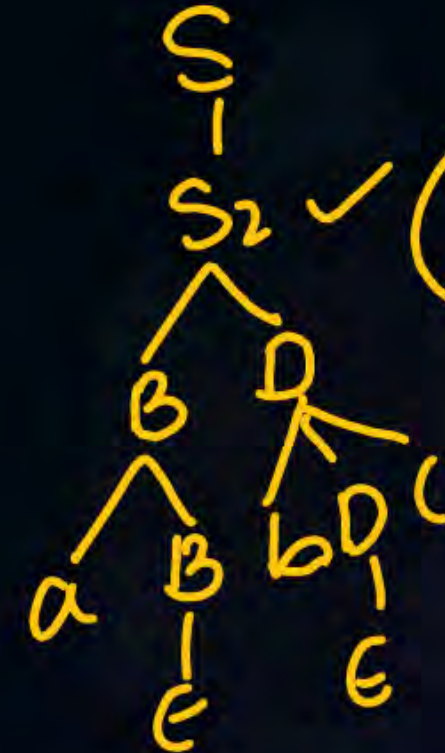
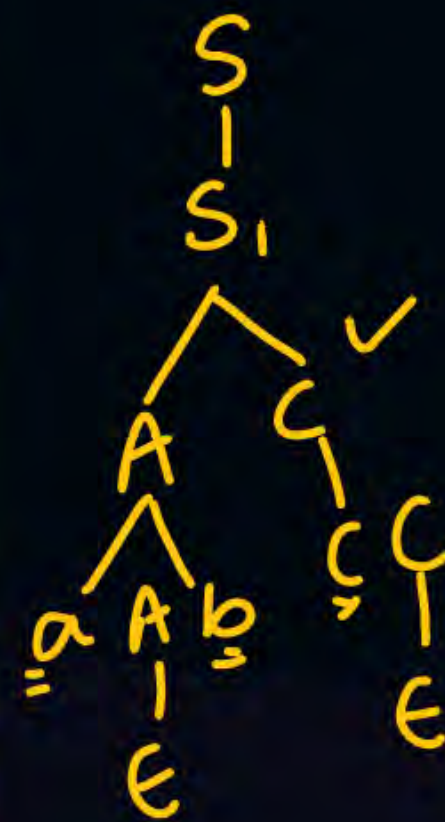
$$A \rightarrow aAb / \epsilon$$

$$C \rightarrow cC / \epsilon$$

$$\checkmark S_2 \rightarrow BD$$

$$B \rightarrow aB / \epsilon$$

$$D \rightarrow bDc / \epsilon$$



abc

$$L = \underline{a^n b^n c^m d^m} \cup \underline{a^n b^m c^m d^n}$$

a) ambig b) unambig

1) $L = L_1 \cup L_2$

2) Not collapsing

3) $L_1 \cap L_2 = \underline{a^n b^n c^n d^n} = \infty \text{ Set}$

$$S \rightarrow X/Y$$

$$X \rightarrow AB$$

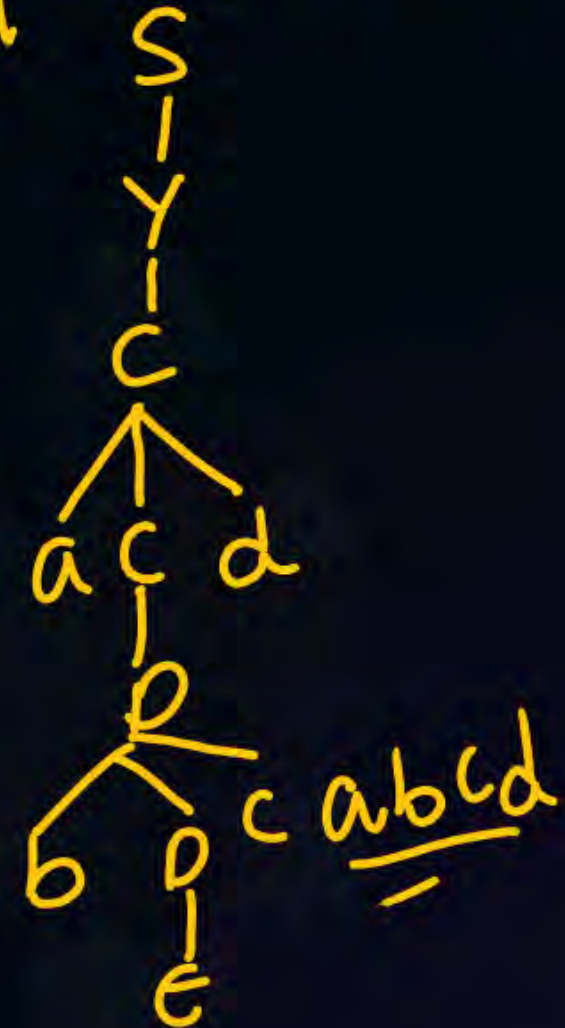
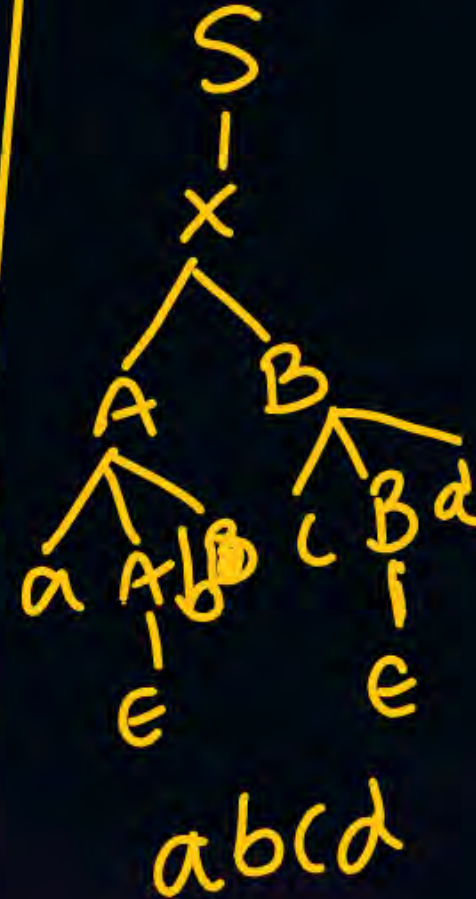
$$A \rightarrow aAb/\epsilon$$

$$B \rightarrow cBd/\epsilon$$

$$Y \rightarrow C$$

$$C \rightarrow aCd/D$$

$$D \rightarrow bDc/\epsilon$$



$$L = \{a^m b^n / m > n\} \cup \{a^m b^n / m < n\}$$

a) ambⁱ b) $\neg$ ambⁱ

1) $L = L_1 \cup L_2$

2) collapse ✓

$$\underline{a^m b^n / m \neq n} \checkmark$$

$\neg$ ambⁱ language

$$L = a^m b^n / \underline{m > n} \cup a^m b^n / \underline{m < n} \cup a^m b^n / \underline{m = n}$$

a) ambi b) unambi

1) $L = L_1 \cup L_2 \cup L_3$

2) collapse $\rightarrow \underline{\underline{a^* b^*}} \rightarrow$ unambiguous language

→ If all grammars of a language are ambiguous then
Then language is ambiguous.



THANK - YOU